Keeping subgroup identities in a coset action

Human Compression project

Algorithm 3.1 of Leemans and Rodrigues [1] takes an arbitrary group as input. It passes maximal subgroups to a coset action and then takes their full inverse images. This operation can replace a maximal subgroup by the whole group. We give the exact obstruction and a repair that leaves the orbit computation unchanged. The issue does not arise for the nonabelian simple groups in the source's numerical tables.

1 The inverse image of a subgroup image

Let G be finite, let H be a maximal proper subgroup, and let $\varphi : G \longrightarrow \mathrm{Sym}(H\backslash G)$ be the action on right cosets by right multiplication. Its kernel is

$$N = \mathrm{core}_G(H) = \bigcap_{g \in G} g^{-1}Hg.$$

In particular this action need not be faithful, contrary to the unqualified assertion on source page 4.

Proposition 1. *For every subgroup $K \leq G$, $\varphi^{-1}(\varphi(K)) = KN$. If K is maximal proper, this inverse image equals K when $N \leq K$, and equals G otherwise. For a point $x = Hg$, however,*

$$\varphi^{-1}\big(\varphi(G)_x\big) = g^{-1}Hg. \tag{1}$$

Proof. An element y has image in $\varphi(K)$ exactly when $\varphi(y) = \varphi(k)$ for some $k \in K$, or equivalently $k^{-1}y \in N$. This proves the first equality. Normality of N makes KN a subgroup containing K; maximality gives the stated dichotomy. Finally $\varphi(y)$ fixes Hg exactly when $Hgy = Hg$, equivalently $gyg^{-1} \in H$. This proves (1), with no faithfulness assumption. □

The printed algorithm calls its image subgroup M_2 and outputs $\{\varphi^{-1}(M_2), \varphi^{-1}(G_x)\}$. In this notation, if the original subgroup was K, the actual output is $\{KN, g^{-1}Hg\}$. The separate instruction to compute the orbits of an undefined K is an evident notation error; interpreting it as the just-constructed image subgroup does not repair the inverse-image problem.

2 A counterexample of order four

Take the additive group $G = \mathbb{F}_2^2$ with distinct order-two subgroups $H = \langle(1,0)\rangle$ and $K = \langle(0,1)\rangle$. Both are maximal, and $N = H$. On the two cosets of H, the image of K is transitive. Its one orbit therefore has size two and passes the printed orbit-size test. For either representative point the printed output is

$$\{G, H\},$$

since $KH = G$ and every point stabilizer in the image is trivial. This is not a pair of maximal proper subgroups. The resulting coset incidence graph has one vertex on its G side and two on its H side. Each of the latter has degree one, so the geometry is not firm. By contrast the pair $\{K, H\}$ gives the complete bipartite graph $K_{2,2}$ and is firm.

The small accompanying certificate enumerates the four elements and their three maximal subgroups. Each of the six ordered choices of distinct subgroups produces the same obstruction. It also constructs the two incidence graphs just described. This finite calculation checks the counterexample; Proposition 1 proves the obstruction for all finite groups.

3 The corrected algorithm

Keep the list of original maximal subgroups together with their images. For each original pair H, K in the representative list, compute the $\varphi(K)$-orbits on $H\backslash G$. For a representative x of each orbit of size greater than one, output

$$\{K, \varphi^{-1}(\varphi(G)_x)\}. \tag{2}$$

As in the source, retain only one pair from each orbit of $\mathrm{Aut}(G)$ on unordered pairs. This last step is a required equivalence test, not something supplied by the coset orbit calculation alone.

Theorem 2. *The corrected algorithm produces precisely the unordered pairs of distinct maximal proper subgroups, up to the equivalence specified in the source. Consequently it classifies the source's firm, connected, flag-transitive primitive coset geometries of rank two for every finite group, subject to a complete maximal-subgroup list and an exact automorphism-equivalence test.*

Proof. For $x = Hg$, put $H_x = g^{-1}Hg$. By (1) this is the second subgroup in (2), so both output subgroups are maximal proper. The orbit size is

$$|x^K| = [K : K \cap H_x].$$

It equals one exactly when $K \leq H_x$, which by maximality is equivalent to $K = H_x$. Thus the orbit-size condition retains exactly the distinct pairs. Such a pair generates G: a subgroup containing both would strictly contain a maximal subgroup. Both residue sizes $[K : K \cap H_x]$ and $[H_x : K \cap H_x]$ are at least two, proving firmness. Generation gives connectedness, and right multiplication is transitive on incident pairs of cosets: any element in their intersection carries the standard incident pair to that pair.

For completeness, start with any ordered pair A, B of distinct maximal subgroups. Conjugate it so that B becomes a chosen representative K. The resulting first subgroup is $g^{-1}Hg$ for some chosen representative H and some $g \in G$. The point Hg lies in a K-orbit whose chosen representative is Hgk for some $k \in K$. Its stabilizer is $k^{-1}g^{-1}Hgk$. Conjugation by k fixes K and carries the previous pair to the produced pair. The orbit is nonsingleton because the subgroups are distinct. Hence every pair is produced up to inner conjugacy; the final exact test takes the required quotient by all automorphisms and type interchange. □

If G is nonabelian simple, every proper maximal subgroup has trivial core. Proposition 1 then shows that the printed inverse-image step already returns K. The correction above therefore does not explain discrepancies in the sporadic numerical tables; those are separate source claims.

The subgroup-orbit method, the orbit-size test and the final automorphism test originate in [1]. The kernel identity and the orbit-stabilizer theorem are elementary group theory. The contribution here is the diagnosis of the stated arbitrary-group scope, its explicit counterexample, and the correction retaining the original subgroup. No new classification of the sporadic incidence graphs, Lean proof or compression measurement is claimed by this component.

References

[1] D. Leemans and B. G. Rodrigues, *On primitive geometries of rank two*, The Art of Discrete and Applied Mathematics 2 (2019), P1.08, doi:10.26493/2590-9770.1301.165, pages 4–5, Algorithm 3.1.